MATH 15200: Calculus II
Homework 5: Inverse Functions and Techniques of Integration

Due: Monday, November 16, 2026, at 11:59 PM.

This is the written portion of Homework 5; complete the routine online portion separately in Edfinity. Upload your written solutions to Gradescope, via Canvas. You may collaborate, but you must write and submit your own solutions independently.

1. Let $f(x) = x + e^x$. Since $f'(x) = 1 + e^x > 0$, the function f is increasing, so it has an inverse function f^{-1}. There is no formula for f^{-1} in terms of familiar functions. Nevertheless, evaluate

$$\int_1^{1+e} f^{-1}(y)\,dy.$$

Hint: First show that

$$\int_{f(a)}^{f(b)} f^{-1}(y)\,dy = b\,f(b) - a\,f(a) - \int_a^b f(x)\,dx.$$

To do this, make the substitution $y = f(x)$ on the left side, and then integrate by parts. Then find a and b with $f(a) = 1$ and $f(b) = 1 + e$.

2. For all $x \neq 0$, set

$$g(x) = \arctan x + \arctan\Big(\frac{1}{x}\Big).$$

Show that $g'(x) = 0$ for every $x \neq 0$. Then show that $g(x) = \frac{\pi}{2}$ for $x > 0$ and $g(x) = -\frac{\pi}{2}$ for $x < 0$. Since g' is zero wherever g is defined, why doesn't this contradict the fact that a function whose derivative is zero at every point of an interval is constant on that interval?

Hint: Once you know that g is constant on $(0, \infty)$ and constant on $(-\infty, 0)$, find each constant by evaluating g at a convenient point of that interval.

3. Let $a, b > 0$. Find the area enclosed by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Check that when $a = b = r$, your answer is the area of a circle of radius r.

Hint: By symmetry, the area is four times the area of the part of the ellipse in the first quadrant. Solve for y on the upper half of the ellipse, set up the integral, and use the substitution $x = a\sin\theta$.